

Social Media (Pie Chart): Grade 9

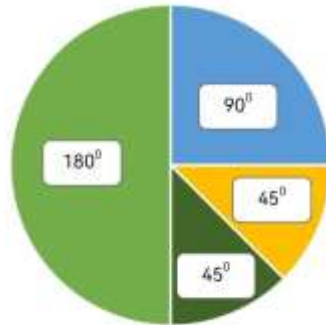
A. Title of card: Pie Charts
B. Competency, Element and Performance Criteria: C 2.4: Represent and interpret data from charts/graphs in daily life Element 1: Read and interpret bar charts, line graphs and pie charts Performance Criteria: Pie charts - Level 1: Construct pie charts & Level 2: Read and interpret data represented in pie charts
C. Purpose of activity: This activity will develop students' understanding of pie charts, enabling them to represent and interpret data from these charts in daily life
D. Learning Outcomes: By the end of the lesson, students should be able to: • convert data into angles for pie charts • accurately draw a pie chart, using a protractor or digital tools (if available) • read and interpret data represented in pie charts
E. Resources and materials: • Coloured whiteboard markers • Protractors • Compasses • Coloured pencils

F. Teaching trajectories/Implementation guidelines:

Activity 1: Review of prerequisites - Fraction and angles in a circle

Steps:

1. Draw a circle on the board and divide it into 4 equal parts. Ask students:
 - If the circle represents a whole, what fraction is 1 part?
 - If one turn is 360° , what angle is each part?
2. Give simple fractions e.g. $\frac{1}{2}$, $\frac{1}{3}$ and ask students to calculate the corresponding angles.



Using the visual chart, ask students:

- If half (50%) of your class uses Instagram, how many degrees does that amount to??
- If three quarter (75%) of your class use TikTok, how many degrees does that amount to?

Activity 2: Time spent on social media Daily

3. Demonstration - Use given data to draw a pie chart using different colours for different segments. Show the students how to label the different segments and choose an appropriate title for the pie chart.

You may survey the class or use the data given below.

Social Media Platform	Snapchat	TikTok	YouTube	Instagram
Time Spent (hours)	2	4	1	2

Calculate the total number, in hours, spent on social media.

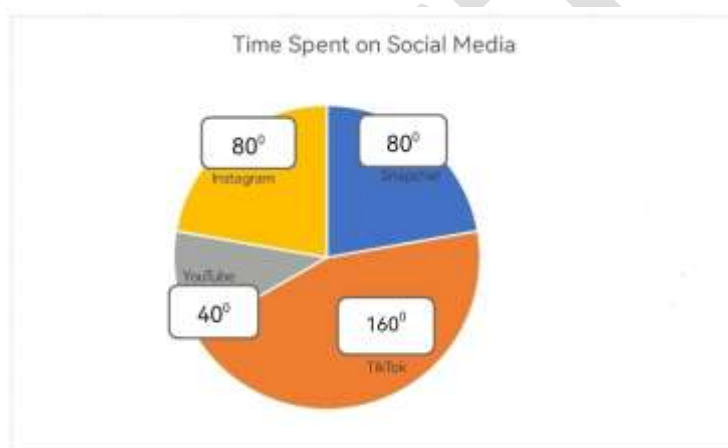
Solution:

Total time spent = 9 hrs

Find the corresponding angles:

Social Media	Time Spent (hrs)	Fraction	Angle
Snapchat	2	$\frac{2}{9}$	$\frac{2}{9} \times 360 = 80^\circ$
TikTok	4	$\frac{4}{9}$	$\frac{4}{9} \times 360 = 160^\circ$
YouTube	1	$\frac{1}{9}$	$\frac{1}{9} \times 360 = 40^\circ$
Instagram	2	$\frac{2}{9}$	$\frac{2}{9} \times 360 = 80^\circ$

Draw the pie chart:



4. Students are asked interpretation questions:

- Which platform has the largest segment? What does that mean?
- Which platform(s) has the smallest segment? What does that mean?

Activity 3: Favourite Social Media Platform

5. In pairs, students use data provided on their favourite social media platforms for a class of 36 students and draw a pie chart using a pair of compasses and a protractor or digital/online tools (if available). Teacher assists struggling students.

The table below shows the favourite social media applications among 36 students.

Social Media Platform	Instagram	TikTok	Facebook	WhatsApp
Number of students	13	11	5	7

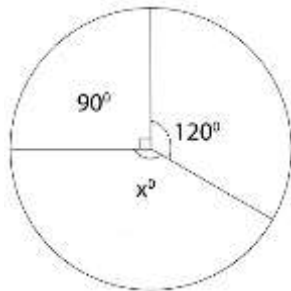
Use a pair of compasses and a protractor to draw a pie chart to represent the above information.

G. Assessment

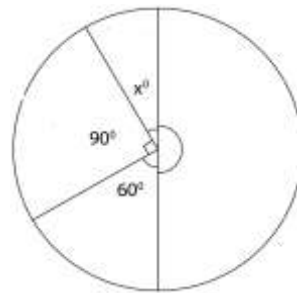
Question 1 – Finding angles in a Pie chart

Find angle x in the pie charts below.

(a)

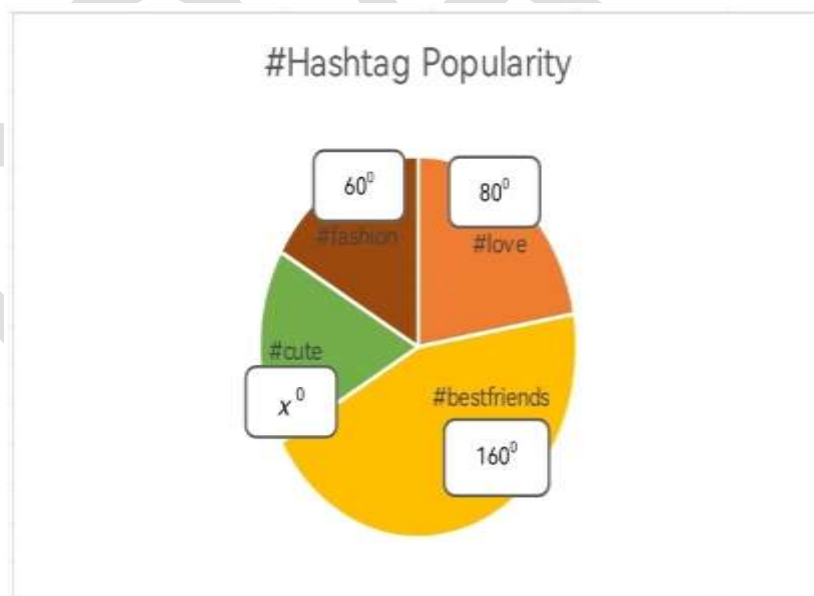


(b)



Question 2 – Interpret a Pie Chart

The pie chart below shows the #hashtag prefer by 90 social media users.



(a) Find the value of x .

(b) What #hashtag is the **most** prefer by social media users?

(c) (i) Find the fraction of users who prefer #love.

(ii) Find the number of users who prefer #love.

Note: (c) may be used as an example

(d) (i) Find the fraction of users who prefer #fashion.

(ii) Find the number of users who prefer fashion.

(e) How many users prefer #bestfriends?

Question 3 – Constructing a Pie Chart

The table below shows the number of likes, shares and comments posted for a Facebook post.

Likes	Shares	Comments
120	80	40

Use a pair of compasses and a protractor to draw a pie chart to represent the above information.

H. Extension of activity